

WEEK 3 DAY 18

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Model & Method Comparison

State of Charge (SOC %) — Full Comparison

Method	Features	MAE(%)	RSME(%)	R^2
Mean baseline	—	23.041381	27.508849	-0.047575
Linear Regression	raw	3.472758	4.585786	0.970888
Linear Regression	engineered	23.193436	27.771279	-0.068143
Random Forest	raw	3.284465	10.474381	0.848121
Random Forest	engineered	22.942698	28.916572	-0.158060
Gradient Boosting	raw	3.300615	10.082012	0.859287
Gradient Boosting	engineered	22.598706	27.527405	-0.049465
MLP (ANN)	engineered	25.932455	31.611177	-0.383945

Remaining Charging Time (s)

- **The Challenge:** Unlike SOC (which spans full operational drive cycles across diverse temperatures), charging time is derived strictly from Status == 'CHA' segments. This yields a far smaller, highly repetitive dataset.
- **Empirical Metric Results:**
 - **MLP (ANN):** MAE = 9,001.75 s (~2.5 hours) | RMSE = 18,410.15 s | $R^2 = -0.160$
- **Why Neural Networks Failed Here:** Because constant-current/constant-voltage (CC-CV) charge cycles follow identical procedures, the multi-layer perceptron memorized the handful of training charge trajectories rather than learning a generalized relationship.
- **The Solution:** We transitioned to a scaled-down Random Forest (200 trees, max depth of 10) to cap model capacity and prevent memorization.

Interpreting MAE, RMSE, and R^2 in Real-World Battery Applications

- **Mean Absolute Error (MAE):** Represents the typical magnitude of error in real units (e.g., 3.28%SOC or seconds remaining). It provides an intuitive, clear measure of average model precision.
- **Root Mean Squared Error (RMSE):** Weights larger errors more heavily due to the squaring penalty. The gap between Random Forest's 3.28%MAE and 10.47%RMSE indicates occasional large mispredictions—specifically at extreme operating boundaries.
- **R^2 Score (Coefficient of Determination):** Measures how much target variance is explained by the model. A score near 1.0 shows strong predictive power, whereas negative values indicate the model performs worse than a simple horizontal mean line

Overfitting: Read Train and Test Together

- **Evaluating the Gap:** Test metrics alone can be misleading. A model might score well by chance, but comparing train vs. test MAE reveals true generalization.
- **Healthy Generalization:** Small gaps between train and test performance (paired with a high R^2) confirm the model learned genuine electrochemical patterns rather than dataset noise.
- **Identifying Failures:**
 - **Overfitting Flag:** A train-test gap exceeding 50% signals that the model overfitted its training files.
 - **Underfitting Flag:** A training $R^2 < 0.60$ shows the model failed to capture the basic underlying trend, calling for higher capacity or cleaner inputs.

Why Both Targets Cannot Be Judged by the Same Standard

- **SOC Target (Rich Signal):** Includes continuous sensor log points spanning 6 ambient temperatures and 14 distinct drive cycles. This provides abundant training signals across the entire state space.
- **Charging Time Target (Sparse Signal):** Contains only one charge segment per file. Because constant-current profiles are highly uniform, the data lacks structural variety.
- **Engineering Impact:** Lower R^2 scores on charging time stem from data scarcity rather than poor model architecture. This directly justifies using restricted-depth decision trees instead of deep neural networks for time estimation.

Measuring Model Capacity Across Different Architectures

- **Linear Models:** Constrained to 1 coefficient per feature plus an intercept. Extremely lightweight and fast, but incapable of capturing nonlinear cell dynamics.
- **Tree Ensembles (Random Forest & Gradient Boosting):** Capacity is determined by $n_estimators \times max_depth$ (e.g., 300 trees $\times$ depth 14). They excel at capturing non-linear behavior but require careful hyperparameter caps to avoid leaf memorization.
- **Artificial Neural Networks (MLP):** Parameter scale is driven by total trainable weights ($64 \rightarrow 32 \rightarrow 1$). While capable of fitting complex curves, they require significantly larger datasets to generalize effectively

Key Physical Regimes Driving Prediction Errors

- **The Low-SOC "Knee" Region:** Near 0% – 15%SOC, the cell voltage drops steeply. Small voltage changes translate to massive SOC variations, causing residual error plots to fan out.
- **Charge Segment Transitions:** Rapid current fluctuations during the transition from Constant Current (CC) to Constant Voltage (CV) modes create sharp discontinuities that tree splits struggle to smooth out.
- **Thermal Extremes:** Data collected at -20°C and 40°C exhibits internal resistance shifts that deviate significantly from room-temperature (25°C) behavior.

Selecting Our Primary Model Candidates

- **State of Charge (SOC) Winner: Random Forest / Gradient Boosting on Raw Features**
 - **Why:** Achieved the lowest overall test error (**MAE = 3.28%**) while maintaining strong variance explanation ($R^2 = 0.85 - 0.97$).
- **Remaining Charge Time Winner: Constrained Random Forest (200 Trees, Depth 10)**
 - **Why:** Successfully avoids the catastrophic overfitting observed in deep neural networks on sparse charge cycle data.
- **Next Steps:** Proceed to hyperparameter optimization on these shortlisted models using strict validation splits—keeping the final test set untouched.

THANK YOU